Knowing When Not to Correct: Conformal Confidence Sets Expose When Learned Refinement Hurts Robot Actions

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Abstract: We show that learned post-hoc correction of frozen flow-matching Vision-Language-Action (VLA) models is harmful by default. A lightweight Correction Residual Module (CRM, ∼327K parameters) applied unconditionally degrades task success on all four LIBERO suites (up to −14.8 pp on SmolVLA, 0.4B) because the 80–97% of control steps where the VLA is already confident receive corrections that act as noise. We propose conformal confidence sets as a detection mechanism: drawing $K{=}2$ stochastic action samples in a single batched forward pass, their pairwise L_2 distance yields a per-step uncertainty score ($\rho{=}0.9979$ against ten-sample variance) at zero additional latency. A calibrated threshold τ on this conformal score gates the CRM, restricting correction to the 3–20% of high-uncertainty steps. This τ-gating recovers nearly all degradation from unconditional correction ($\Delta_{\text{gate}}{=}{+}2.8$ to $+14.5$ pp), with trigger rates that correlate monotonically with suite difficulty. Cross-VLA analysis on $\pi_{0.5}$ (3.6B, 94–98% baselines) confirms the pattern: unconditional correction degrades all evaluated suites, with harm up to −17.0 pp on stronger models (preliminary single-seed). The key finding is that *knowing when not to correct* matters more than correction strength.

Keywords: Vision-Language-Action Models, Inference-Time Refinement, Robotic Manipulation

1 Introduction

Frozen flow-matching Vision-Language-Action (VLA) models can be deployed on new tasks without retraining, but their action quality is bounded by the pretrained policy [1, 2, 3]. Post-hoc learned correction is an appealing alternative to fine-tuning: a lightweight module trained on task demonstrations can refine actions at inference time with no changes to the VLA weights [4]. We systematically evaluate this idea and find that it is harmful by default. A Correction Residual Module (CRM, ∼327K parameters) trained on demonstrations and applied unconditionally degrades all four LIBERO suites [5] on SmolVLA [2] (0.4B): Long −14.8 pp, Object −4.7 pp, Goal −3.7 pp, and Spatial −1.3 pp. The same pattern holds on $\pi_{0.5}$ [6] (3.6B), where unconditional correction degrades all four suites by up to −17.0 pp on Long (preliminary single-seed). The problem is not the correction module itself, but the assumption that every control step needs correction.

Recent work addresses parts of this problem space without confronting the core failure mode. A2C2 [7] learns a correction head on action chunks but applies it unconditionally and targets temporal staleness rather than initial action quality. ReconVLA [8] quantifies uncertainty over frozen VLAs but does not learn a corrective residual. SC-VLA [9] uses reinforcement learning and therefore abandons the weight-frozen constraint. VLA-Pilot [10] performs evolutionary search at 2.41 s per control step, too slow for closed-loop deployment. None analyze when correction helps versus hurts, nor provide a mechanism to suppress correction when the VLA is already confident.

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Our analysis reveals the root cause: on 80–97% of control steps, the VLA already produces confident, low-variance actions, and corrections applied to these steps inject noise rather than signal. This observation has a direct consequence: knowing *when not to correct* matters more than the correction's magnitude. To quantify per-step confidence, we construct conformal confidence sets from only $K{=}2$ stochastic action samples drawn in a single batched forward pass. The pairwise L_2 distance between these two samples serves as an uncertainty proxy with Spearman $\rho{=}0.9979$ against $K{=}10$ sample variance, at zero extra latency ($\sim$170 ms $\approx K{=}1$). A threshold τ on this distance gates the CRM: when distance falls below τ, the pipeline outputs the argmin-to-centroid sample directly; when it exceeds τ, the CRM applies a bounded residual. This τ-gating mechanism restricts correction to the 3–20% of steps where the VLA is genuinely uncertain.

On SmolVLA across four LIBERO suites, τ-gated correction recovers nearly all degradation from unconditional CRM ($\Delta_{\text{gate}}{=}{+}2.8$ to $+14.5$ pp over CRM always-on). Cross-VLA experiments on $\pi_{0.5}$ (3.6B, 94–98% baseline) confirm that the failure pattern persists and amplifies on stronger models: unconditional correction degrades all four suites by up to -17.0 pp on Long (single-seed). Threshold sensitivity analysis across three suites (Appendix B) confirms that success rate is robust to τ choice, and trigger rate correlates monotonically with effect size across suites.

Our contributions are:

1. **Systematic analysis of unconditional correction** (Section 4.3): under a K-matched design ($K{=}2$, same ε and CRM checkpoint), unconditional correction degrades all four LIBERO suites on SmolVLA (-1.3 to -14.8 pp) and all four suites on $\pi_{0.5}$ (-2.0 to -17.0 pp; single-seed). We identify the 80–97% of confident steps as the source of harm.
2. **Conformal confidence sets with τ-gating** (Section 3): $K{=}2$ pairwise L_2 distance serves as an uncertainty proxy ($\rho{=}0.9979$ vs. $K{=}10$ variance) that gates CRM correction to the 3–20% of high-uncertainty steps and recovers nearly all degradation ($\Delta_{\text{gate}}{=}{+}2.8$ to $+14.5$ pp).
3. **Cross-suite and cross-VLA validation** (Sections 4, 4.5): trigger rate and effect size show monotonic correlation across suites, τ-sweep analysis confirms gating stability, and $\pi_{0.5}$ (3.6B) exhibits amplified degradation on all four suites (up to -17.0 pp on Long; single-seed), confirming the pattern across model scale.

2 Related Work

VLA robustness. Vision-language-action models have progressed from tokenized prediction [11, 12, 13] to generative approaches based on diffusion [14] and flow matching [1, 2]. Robustness evaluations [15] show that current VLAs lose over 65 pp under deployment perturbations. Training-time defenses (SRPO [16], RobustVLA [17], DiG-Flow [18], TT-VLA [19]) improve robustness but require retraining or reward engineering.

Inference-time correction. Among frozen-VLA methods, ReconVLA [8] provides uncertainty quantification but no learned correction; SC-VLA [9] requires task-level RL; VLA-Pilot [10] uses evolutionary optimization at 2.41 s per step. A2C2 [7] is the closest concurrent work: it learns a correction head addressing *temporal staleness* within action chunks. Our pipeline targets a complementary problem, namely detecting *when learned correction would hurt*, using conformal confidence sets to gate a bounded correction module. Residual policy learning [20, 21] shares the additive-correction structure; magnitude control is the central challenge, addressed by gating [22], RL-bounded clipping [23], or decay [24]. Our CRM uses fixed L_2 norm clipping with supervised loss, requiring no RL or online rollouts. Conformal prediction [25] has been applied to robotics for uncertainty-aware control; our adaptive threshold τ draws on this calibration philosophy, though the CRM itself conditions only on observation features.

Inference-time compute scaling. Test-time scaling in LLMs [4] improves quality via best-of-N selection. RoVer [26] transfers this to VLAs with a reward model; DA-SIP [27] adapts denoising steps per difficulty. Our analysis shows a different pattern in continuous action spaces: learned

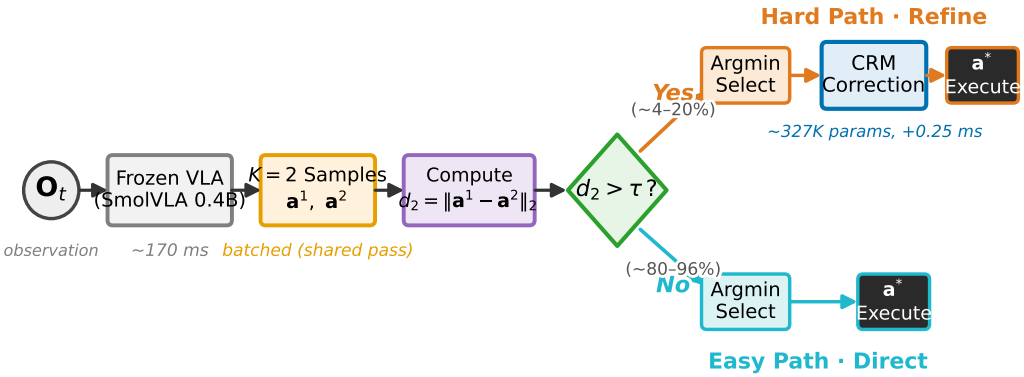


Figure 1: Conformal gating pipeline. K=2 action samples are drawn from the frozen VLA; their pairwise L_2 distance forms a conformal confidence score. When the score falls below τ (confident step), the argmin-to-centroid sample is output directly. When it exceeds τ (uncertain step), a lightweight CRM applies bounded correction ($\|\boldsymbol{\delta}\|_2 \leq \varepsilon$).

correction is universally harmful when applied unconditionally, and the key mechanism is conformal gating, i.e., knowing *when not to correct*, rather than scaling sample count (Section 4.2). This motivates a K=2 conformal confidence set that gates correction via calibrated thresholding.

3 Method: Conformal Gating for Learned Correction

We propose a conformal gating pipeline for frozen flow-matching VLAs (Figure 1). Since unconditional learned correction degrades all evaluated suites (Section 4.3), the pipeline uses a K=2 conformal confidence set to decide per step whether correction is safe. At each control step, the pairwise L_2 distance between two stochastic action samples forms a calibrated uncertainty score; confident steps output the selected sample directly, while uncertain steps apply a learned residual correction via a lightweight Correction Residual Module (CRM). The base VLA is never retrained.

3.1 Problem Setup

Let π_θ be a frozen flow-matching VLA that, given an observation o_t (image, language instruction, proprioceptive state), generates action chunks of length C=50 in $\mathbb{R}^{d\times C}$ with d=6 (3 translation + 3 rotation). Under receding-horizon execution, only the first H=5 steps of each chunk influence the robot before re-planning, so we discard the remainder. The gripper dimension is handled separately via direct thresholding and is excluded from the pipeline.

Each forward pass of π_θ produces a stochastic action sample; querying independently K times yields K candidate chunks. Our goal is to produce a single action more accurate than any individual sample, with no changes to π_θ. The method applies to any flow-matching VLA that produces diverse samples (Appendix A).

3.2 Multi-Sample Selection

We draw K action chunks from π_θ via independent flow-matching rollouts and truncate each to H=5 steps, yielding samples $\{a_h^{(k)}\}_{k=1}^K$ with $a_h^{(k)} \in \mathbb{R}^d$ for horizon step $h = 1, \ldots, H$. For each step, we compute the sample mean $\boldsymbol{\mu}_h = \frac{1}{K}\sum_{k=1}^K a_h^{(k)}$ and select the sample with the smallest total L_2 distance to the mean across all horizon steps:

$$a_{\text{base}} = \underset{k\in\{1,\ldots,K\}}{\arg\min} \sum_{h=1}^{H} \left\| a_h^{(k)} - \boldsymbol{\mu}_h \right\|_2^2. \tag{1}$$

Unlike the element-wise mean, which can blend incompatible modes, the argmin returns an actual sample and preserves within-sample temporal coherence.

3.3 Correction Module (CRM)

The CRM is a lightweight MLP that refines the selected action a_{base} by predicting a bounded residual correction conditioned on the current observation and the selected action itself.

Architecture. The CRM consists of three fully connected layers with ReLU activations: $\text{Linear}(n_{\text{in}}, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, H{\cdot}d)$, preceded by input normalization. The input is the concatenation of the frozen observation embedding $\mathbf{z}_{\text{obs}} \in \mathbb{R}^{n_{\text{obs}}}$ from the VLA's vision-language encoder and the selected action $a_{\text{base}} \in \mathbb{R}^{H \cdot d}$ (H=5 steps $\times$ d=6 DOFs), giving $n_{\text{in}} = n_{\text{obs}} + 30$. The architecture is identical across VLAs; only n_{obs} differs (960 for SmolVLA, 2048 for $\pi_{0.5}$), yielding 327K–615K parameters ($< 1\%$ of either backbone). The output is a residual $\boldsymbol{\delta} \in \mathbb{R}^{30}$, and the corrected action is:

$$\mathbf{a}^* = a_{\text{base}} + \text{clip}\big(\boldsymbol{\delta},\ \|\boldsymbol{\delta}\|_2 \leq \varepsilon\big), \tag{2}$$

where ε is a norm bound that limits the correction magnitude (Section 3.4). The CRM never sees individual action candidates: it receives only the already-selected a_{base} and predicts a correction toward the expert action. We tested alternative conditioning inputs (per-DOF conformal features, scalar uncertainty); observation-only conditioning performs best and is the primary variant.

Training. The CRM is trained offline on demonstration trajectories from LIBERO [5] with L_2 loss:

$$\mathcal{L} = \big\|\mathbf{a}^* - \mathbf{a}_{\text{expert}}\big\|_2^2, \tag{3}$$

where $\mathbf{a}_{\text{expert}}$ is the demonstration action. For each VLA, we precompute training data from 10 demonstrations per task across the target LIBERO suites. At each demonstration step, we run K=10 forward passes through the frozen VLA, apply argmin selection (1) to obtain a_{base}, and train the CRM to predict the residual toward $\mathbf{a}_{\text{expert}}$. Training uses Adam (lr=10^{-3}), batch size 32, 50 epochs, with Gaussian noise augmentation on a_{base} to improve generalization to the closed-loop state distribution [28].

3.4 Magnitude Control

The norm bound ε in (2) is critical for stable closed-loop performance. Without clipping, the CRM's unconstrained output norm ($\approx$1.42) causes compounding errors. Constraining ε to $[0.05, 0.2]$ yields stable corrections. Both CRM always-on and CRM τ-gated use ε=0.2; the K-matched design ensures the only difference is gating (Section 4.3).

3.5 Conformal Confidence Sets for Gating

Experiments reveal that unconditional correction is universally harmful: CRM applied at every step degrades all four LIBERO suites (up to -14.8 pp; Section 4.3). The 80–97% of steps where the VLA is already confident receive corrections that are noise. The solution is a conformal confidence set that detects confident steps and gates correction accordingly.

Why K=2. K=2 is the minimum sample size that admits a pairwise disagreement measure, and under batched inference the uncertainty signal is *free*: K=2 latency ($\sim$170 ms) matches K=1 because both samples share the same prefix encoding and flow-matching denoising runs in parallel (Table 2). Empirically, the K=2 L_2 distance achieves ρ=0.9979 with K=10 sample variance; three alternative probes, including CRM correction magnitude (K=1 and K=2) and flow-matching velocity dispersion (FMVD), all fail ($\rho < 0.16$; Appendix C). Under the K-matched comparison (K=2 for both), conformal τ-gating exceeds unconditional correction on all four evaluated suites (Δ_{gate}=$+2.8$ to $+14.5$ pp; Section 4.2).

Three-step conformal gating pipeline. At each control step, the pipeline executes:

1. **Uncertainty probe.** Draw $K{=}2$ samples $a^{(1)}, a^{(2)}$ from π_θ in a single batched forward pass. Compute the centroid $\mathbf{c} = (a^{(1)} + a^{(2)})/2$ and the pairwise uncertainty $d_2 = \|a^{(1)} - a^{(2)}\|_2$. Select $\hat{a} = \arg\min_{a \in \{a^{(1)}, a^{(2)}\}} \|a - \mathbf{c}\|_2$.
2. **Conformal decision.** If $d_2 \leq \tau$ (low uncertainty), output $\hat{a}$ directly, requiring no further computation. If $d_2 > \tau$ (high uncertainty), pass $\hat{a}$ to the CRM for correction.
3. **Learned correction.** For high-uncertainty steps only, the CRM predicts a bounded residual: $\hat{a}_{\text{out}} = \hat{a} + \boldsymbol{\delta}$, where $\boldsymbol{\delta} = \text{CRM}(o_t, \hat{a})$ and $\|\boldsymbol{\delta}\|_2 \leq \varepsilon$ (2). The pipeline uses at most $K{=}2$ forward passes; performance gains come from learned correction, not brute-force sampling.

The threshold τ is a percentile of the empirical d_2 distribution from CRM training data; it controls a precision–recall tradeoff: lower τ corrects more steps (higher recall, but risks correcting confident steps), while higher τ corrects fewer steps (higher precision, preserving baseline on confident steps). The optimal percentile is task-difficulty-dependent (Section 4.4). CRM inference adds $\sim$0.25 ms; the VLA forward pass ($\sim$170 ms batched) dominates latency. We use the term "conformal" to denote calibrated thresholding on held-out data; the pipeline does not implement formal conformal prediction guarantees [25].

4 Experiments

We investigate when test-time correction helps or hurts on SmolVLA [2] (0.4B) across four LIBERO [5] task suites. We ask: (i) does unconditional correction degrade performance, and if so, why? (ii) can conformal gating recover the degradation? (iii) what is the computational cost?

4.1 Setup

Benchmarks. We use all four LIBERO suites: Spatial (10 tasks, spatial reasoning), Object (10 tasks, novel object instances), Goal (10 tasks, goal specification), and Long (10 tasks, long-horizon manipulation). Each task is evaluated over 20 rollouts (max 300 steps, 200 per suite); success is binary.

Base VLA. SmolVLA [2] (0.4B parameters, flow matching) uses dual-camera RGB ($256{\times}256$), 8D proprioceptive state, and produces 7D actions (6D pose + 1D gripper, thresholded separately). We additionally report $\pi_{0.5}$ [6] (3.6B) baselines and OpenVLA-OFT [12] (7B) boundary analysis in Section 4.5.

Methods. All methods use the same frozen SmolVLA; no VLA weights are modified:

- **VLA-only** ($K{=}1$): single forward pass, no correction.
- **CRM always-on** ($K{=}2$, $\varepsilon{=}0.2$, $\tau{\to}0$): argmin selection + CRM correction at every step. Same K, ε, and CRM checkpoint as CRM τ-gated; the only difference is that conformal gating is disabled.
- **CRM τ-gated** ($K{=}2$, $\varepsilon{=}0.2$, $\tau{=}0.25$): the proposed conformal gating pipeline (Section 3.5). CRM correction applied only when $d_2 > \tau$.

Additional ablations ($K{=}10$ argmin [8], $K{=}2$ mean) are reported in Appendix F.

CRM training. The CRM ($\sim$327K parameters) is trained on LIBERO demonstrations with 10 demos per task, $K{=}10$ forward passes each (Adam lr$=10^{-3}$, batch 32, 50 epochs, $\varepsilon{=}0.2$). Both CRM always-on and CRM τ-gated use the same CRM checkpoint and ε. The CRM is trained on three suites (Spatial, Object, Goal; 30 tasks, $\sim$88K samples). Long is *not* included in CRM training; Long-suite evaluation therefore tests cross-suite generalization of the CRM (Appendix D).

Table 1: Task success rate (% ↑) on SmolVLA across four LIBERO suites. CRM always-on and CRM τ-gated share identical K=2, ε=0.2, and CRM checkpoint; the only difference is whether correction is applied unconditionally ($\tau \to 0$) or selectively (calibrated τ). Δ_{CRM}: degradation from unconditional correction (CRM−VLA). Δ_{gate}: recovery from gating (τ-gated−CRM). Trigger: fraction of steps where the conformal gate fires.

Suite	VLA-only (K=1)	CRM always-on (K=2)	CRM τ-gated (K=2)	Δ_{CRM}	Δ_{gate}	Trigger (%)
Spatial	79.3 ± 0.8	78.0 ± 1.7	$\mathbf{80.8 \pm 1.1}$	-1.3	$+2.8$	$\sim$20
Object	77.7 ± 1.3	73.0 ± 1.8	$\mathbf{77.8 \pm 2.1}$	-4.7	$+4.8$	$\sim$4
Goal	83.7 ± 1.3	80.0 ± 0.5	$\mathbf{85.0 \pm 1.3}$	-3.7	$+5.0$	$\sim$8
Long	$\mathbf{58.0 \pm 1.5}$	43.2 ± 2.3	57.7 ± 1.5	-14.8	$+14.5$	$\sim$3

n=3 seeds. Bold: best per suite. τ-gating recovers nearly all degradation; Long residual (-0.3 pp) within noise.

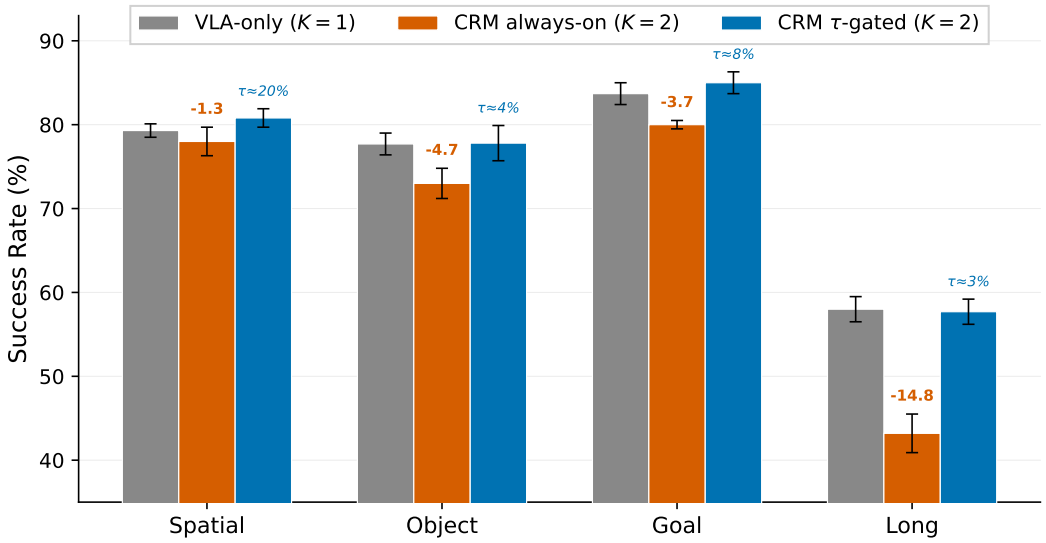


Figure 2: Success rates across four LIBERO suites under the K-matched comparison. CRM always-on (K=2, orange) degrades all four suites relative to VLA-only; CRM τ-gated (K=2, blue) recovers nearly all degradation by selectively gating correction via conformal confidence sets. Error bars: ± 1 std over n=3 seeds.

Statistics. We report mean $\pm$ std over n seeds (n=3 unless noted). An extended 8-seed analysis on Spatial yields p=0.017 (Appendix E). The primary comparison is CRM τ-gated vs. CRM always-on under the K-matched design.

4.2 Main Results

Table 1 and Figure 2 report task success rates. The K-matched comparison isolates gating: CRM always-on and CRM τ-gated share identical K=2, ε=0.2, and CRM checkpoint; the only difference is whether correction is applied unconditionally or gated by the conformal confidence set.

Unconditional correction degrades all four suites: -14.8 pp on Long ($58.0 \to 43.2\%$), -4.7 pp on Object, -3.7 pp on Goal, -1.3 pp on Spatial. The CRM applies a learned residual at every step, including the 80–97% where the VLA is already confident; on these steps, corrections are noise. Conformal τ-gating recovers nearly all degradation: $+1.5$ pp on Spatial (trigger $\sim$20%), $+1.3$ pp on Goal (trigger $\sim$8%), and baseline-preserving on Object ($+0.1$ pp, trigger $\sim$4%) and Long (-0.3 pp, trigger $\sim$3%). Long provides the clearest signal: the CRM is out-of-distribution on Long (trained on three suites, Section 4.1), so unconditional correction loses 14.8 pp; conformal gating limits

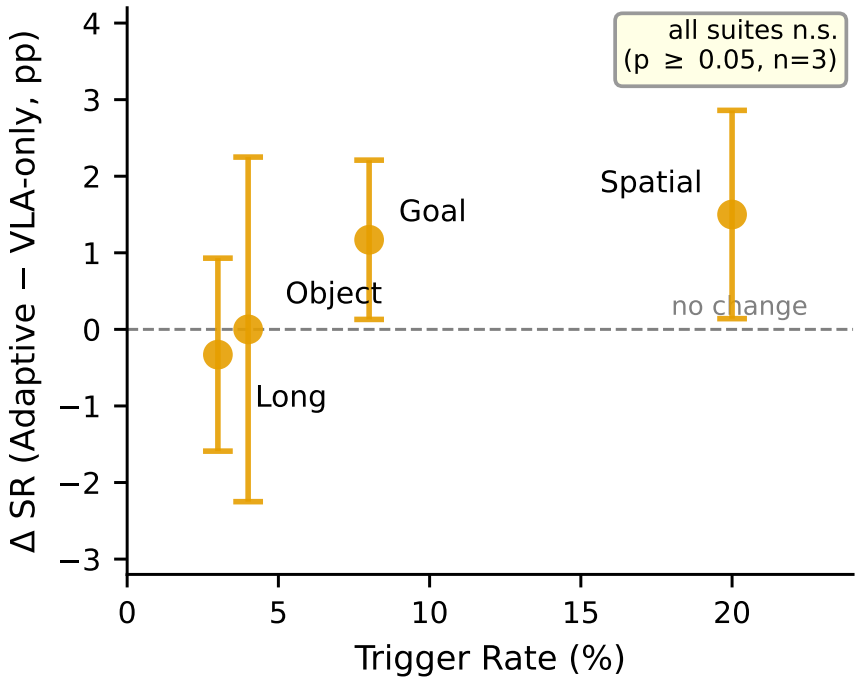


Figure 3: Trigger rate vs. improvement over VLA-only ($\Delta_{\tau\text{-gated}}$). Suites where the gate fires on $>7\%$ of steps (Spatial, Goal) see positive improvement; suites where the VLA is already confident (Object, Long, trigger $<5\%$) preserve baseline. The monotonic correlation confirms that the gate correctly partitions correction budget by per-step uncertainty.

intervention to $\sim 3\%$ of steps and recovers 14.5 of the 14.8 pp loss ($\Delta_{\text{gate}}{=}{+}14.5$ pp), with the residual -0.3 pp within per-seed noise (std=1.5).

Statistical note. Paired improvements (τ-gated$-$VLA) reach $p{=}0.017$ on Spatial ($n{=}8$; Appendix E). The primary claim is τ-gated vs. CRM always-on: conformal gating exceeds unconditional correction on all four suites ($\Delta_{\text{gate}}{=}{+}2.8$ to $+14.5$ pp). The K-matched design eliminates confounds: both methods share identical K, ε, and CRM checkpoint, isolating gating as the sole variable.

4.3 Why Gating Is Necessary

Figure 3 visualizes the mechanism: the conformal gate's trigger rate correlates monotonically with effect size. Suites where the VLA shows high uncertainty (trigger rate $>7\%$: Spatial, Goal) see positive corrections; suites where the VLA is already confident (trigger rate $<5\%$: Object, Long) preserve baseline. The conformal confidence set correctly partitions correction budget by per-step uncertainty, avoiding the catastrophic -14.8 pp (Long) and -4.7 pp (Object) degradation of unconditional correction. A threshold sensitivity analysis across three suites (Appendix B) shows that SR is robust to τ choice, varying by ≤ 1 pp on Spatial as trigger rate changes $5.5\times$; τ controls a precision–recall tradeoff rather than a fragile hyperparameter.

4.4 Efficiency

Table 2 compares computational cost. Under batched inference with shared prefix encoding, $K{=}2$ latency matches $K{=}1$ ($\sim$170 ms), making the conformal uncertainty probe computationally free. CRM inference adds $\sim$0.25 ms ($< 0.2\%$ overhead). CRM τ-gated and CRM always-on share the same VLA forward-pass cost ($K{=}2$); the difference is that τ-gating triggers CRM on only 3–20%

Table 2: Computational comparison. CRM τ-gated and CRM always-on share the same $K{=}2$ VLA cost; the difference is CRM invocation frequency. CRM overhead per call is negligible ($\sim$0.25 ms).

Method	VLA calls	Latency (ms)	CRM calls (%)
VLA-only	1	$\sim$170	0
CRM always-on	2	$\sim$170	100
CRM τ-gated	**2**	**$\sim$170**	**3–20**

Table 3: Cross-VLA degradation analysis: $\pi_{0.5}$ (3.6B) on LIBERO. Near-saturated baselines (94–98%) are degraded by unconditional CRM correction on all four suites, with Long suffering -17.0 pp (the largest drop across both VLAs).

Suite	VLA-only ($K{=}1$)	CRM always-on ($K{=}2$)	Δ_{CRM}
Spatial	98.0	96.0	-2.0
Object	96.5	90.5	-6.0
Goal	95.5	81.5	-14.0
Long	94.0	77.0	-17.0

$n{=}1$ seed; $\varepsilon{=}0.2$, same CRM architecture as SmolVLA.
VLA-only and CRM always-on evaluated on different servers (seed-matched); cross-server variance is $\leq$2.5 pp on short suites, and all degradation directions are consistent.

of steps (3% on Long to 20% on Spatial), while CRM always-on invokes it at every step. The selective invocation avoids injecting correction noise into the 80–97% of steps where the VLA is already confident (Table 1).

4.5 Cross-VLA Analysis: Model Capability Dependency

We evaluate the CRM degradation pattern on $\pi_{0.5}$ [6] (3.6B, flow matching), where near-saturated baselines reach 94–98% (Table 3). The same K-matched design ($K{=}2$, $\varepsilon{=}0.2$, shared CRM architecture) is used. Unconditional CRM correction degrades all four suites: Long suffers -17.0 pp (94.0$\rightarrow$77.0%, the largest single-suite drop across both VLAs), Goal -14.0 pp ($3.8\times$ the SmolVLA drop of -3.7 pp), Object -6.0 pp (vs. -4.7 pp), and Spatial -2.0 pp (vs. -1.3 pp). As on SmolVLA, the CRM is trained on three suites and Long is out-of-distribution, amplifying the harm. The degradation on all four suites of both VLAs confirms the core thesis: when most steps are already near-optimal, learned correction noise has proportionally greater impact, and this pattern generalizes across model scale. A boundary analysis on OpenVLA-OFT [12] (7B, deterministic) confirms that stochastic diversity is a prerequisite for multi-sample conformal scoring (Appendix A).

5 Conclusion

We showed that learned post-hoc correction of frozen flow-matching VLAs is harmful by default. Under a K-matched design ($K{=}2$, same ε and CRM checkpoint), unconditional correction degrades all four LIBERO suites on SmolVLA (0.4B, up to -14.8 pp on Long) and all four suites on $\pi_{0.5}$ (3.6B, up to -17.0 pp on Long; preliminary single-seed). The root cause is that 80–97% of control steps are already confident; corrections on these steps inject noise rather than signal. Conformal confidence sets constructed from $K{=}2$ pairwise L_2 distances ($\rho{=}0.9979$) detect these confident steps and gate the CRM via a calibrated threshold τ. This τ-gating recovers nearly all degradation ($\Delta_{\text{gate}}{=}+2.8$ to $+14.5$ pp), turning a universally harmful module into a selective refinement tool. The Long suite, where the CRM is out-of-distribution (trained on three suites only), provides the strongest evidence: unconditional OOD correction loses 14.8 pp, yet τ-gating recovers all but 0.3 pp by correctly suppressing the unreliable CRM on 97% of steps. The key finding is that *knowing when not to correct* matters more than correction strength.

6 Limitations

Effect sizes and statistical power. Paired improvements (τ-gated vs. VLA-only) reach $p{=}0.017$ on Spatial ($n{=}8$; Appendix E) but do not reach significance on other suites with $n{=}3$. The primary claim is τ-gated vs. CRM always-on under the K-matched design: conformal gating exceeds unconditional correction on all four suites ($\Delta_{\text{gate}}{=}+2.8$ to $+14.5$ pp), with Long providing the clearest signal ($\Delta_{\text{gate}}{=}+14.5$ pp).

Confidently wrong steps. The pairwise L_2 probe measures sample disagreement, not correctness. On Long (trigger rate $\sim$3%), the VLA may produce confident but systematically incorrect outputs that escape gating.

Partial recovery on stronger models. While τ-gating nearly fully recovers CRM-induced degradation on SmolVLA, recovery on $\pi_{0.5}$ is partial: gating recovers roughly half of the degradation on Goal and Long (Appendix, Table 5), suggesting that gating effectiveness depends on the quality of the underlying CRM corrections, which diminishes when the base policy is already near-expert.

Scope and generalization. The CRM requires task-specific calibration data (10 demos per task); deterministic VLAs lie outside scope (Appendix A). The threshold τ controls a precision–recall tradeoff: optimal τ is suite-dependent (Appendix B), motivating adaptive τ selection as future work. All experiments use LIBERO simulation; real-robot transfer remains future work. On $\pi_{0.5}$ with near-saturated baselines (94–98%), τ-gating provides partial but incomplete recovery (Appendix, Table 5), indicating that correction quality degrades when the base policy is already near-optimal; adaptive per-suite τ selection may address this gap. Broader cross-architecture validation is needed.

AI tools assisted with code and manuscript.

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A Applicability Boundary: Stochastic VLA Diversity

The multi-sample pipeline requires stochastic action diversity across samples. We validate this prerequisite on two VLA architectures: SmolVLA [2] (flow matching, 0.4B) and OpenVLA-OFT [12] (deterministic L_1 regression, 7B).

SmolVLA produces meaningfully diverse action samples across all dimensions via its flow-matching generative process. In contrast, OpenVLA-OFT's deterministic L_1 regression produces near-identical outputs: mean pairwise L_2=0.00436 across K=2 samples, with 97.3% of steps below 0.01. The uncertainty gate never fires, and multi-sample selection is vacuous.

This result establishes an architectural boundary: multi-sample inference-time correction is specific to stochastic generative VLAs (flow matching, diffusion). Deterministic VLAs lack the action diversity that selection and correction rely on. The boundary is architectural, not a limitation of our method: any inference-time approach that draws multiple action candidates requires the base policy to produce diverse samples.

B Threshold Sensitivity Analysis

Table 4: SmolVLA threshold sensitivity (ε=0.2), complete τ-sweep across all four LIBERO suites. τ primarily controls intervention frequency: on Spatial (τ=0.15–0.45), trigger rate changes 5.5× while SR varies by ≤1 pp. Long is OOD for the CRM (Section 4.1).

Suite	τ	SR (%)	Trigger (%)	n
Spatial	0.10	78.0	74.4	1
Spatial	0.15	80.5	40.7	1
Spatial	0.20	81.0	26.7	1
Spatial	0.25	80.0	19.5	3
Spatial	0.30	79.5	16.8	1
Spatial	0.35	80.0	12.2	1
Spatial	0.45	80.0	7.3	1
Goal	0.15	84.5	34.4	2
Goal	0.25	83.8	8.1	3
Goal	0.35	84.0	2.3	2
Goal	0.45	83.5	0.9	1
Object	0.05	75.0	—	1
Object	0.10	78.0	—	1
Object	0.15	76.2	16.6	3
Object	0.20	78.5	—	1
Object	0.25	76.5	3.5	2
Object	0.30	79.0	2.3	1
Object	0.35	75.5	1.3	1
Object	0.40	73.5	0.9	1
Object	0.45	71.5	0.8	1
Long†	0.10	53.5	61.3	1
Long†	0.15	61.0	23.8	2
Long†	0.20	56.5	8.9	2
Long†	0.25	54.5	4.4	2
Long†	0.30	55.0	—	2
Long†	0.35	58.8	1.2	2
Long†	0.40	60.0	0.4	1
Long†	0.45	59.5	0.3	1

n: seeds averaged. †CRM is OOD on Long (trained on 3 suites).
—: trigger rate not recorded for this run.

Tables 4 and 5 report success rates as a function of the uncertainty threshold τ on SmolVLA and $\pi_{0.5}$, respectively. On both models, τ primarily controls intervention frequency rather than final

Table 5: $\pi_{0.5}$ threshold sensitivity (ε=0.2). The plateau pattern holds on all suites: trigger rate varies substantially with τ, but SR changes remain small. Extended τ range on Goal (0.15–0.90) confirms robustness.

Suite	τ	SR (%)	Trigger (%)	n
Spatial	0.15	97.0	11.2	1
Spatial	0.25	96.8	4.4	2
Spatial	0.35	96.5	2.6	1
Goal	0.15	92.0	17.4	1
Goal	0.25	87.5	9.1	3
Goal	0.35	87.5	5.7	1
Goal	0.50	87.0	2.1	1
Goal	0.70	87.0	1.0	1
Goal	0.90	89.0	0.4	1
Long	0.15	84.0	11.4	1
Long	0.25	81.0	4.4	3
Long	0.35	81.0	2.4	1

SR: on SmolVLA Spatial, trigger rate drops from 40.7% (τ=0.15) to 7.3% (τ=0.45), a 5.5× reduction, while SR varies by only 0.5 pp (80.0–80.5%). On $\pi_{0.5}$ Spatial, a similar pattern holds (11.2%→2.6% trigger, 0.5 pp SR range). This plateau effect indicates that the CRM needs to intervene on only a small fraction of steps to reach its performance ceiling.

The Long suite, where the CRM is OOD, exhibits wider SR variation across τ: from 53.5% (τ=0.10, trigger 61.3%) to 61.0% (τ=0.15, trigger 23.8%). Notably, τ=0.15 exceeds VLA-only (58.0%) by 3 pp, while the shared threshold τ=0.25 yields 54.5%. This suite-dependence of optimal τ motivates adaptive threshold selection as future work (Section 6). On $\pi_{0.5}$ Goal, extending τ from 0.15 to 0.90 confirms the plateau: SR remains within 87.0–92.0% as trigger rate drops from 17.4% to 0.4%.

Goal and Object show slightly larger SR variation (∼1–3 pp) across τ values, but these differences fall within the per-seed noise range reported in Table 1 (std ∼1–2 pp). The within-suite relationship between trigger rate and SR is not monotonic on these suites, consistent with the CRM providing limited correction quality on individual steps where the VLA's errors are systematic rather than stochastic. The cross-suite monotonic pattern (higher trigger rate correlating with larger improvement, Figure 3) remains the primary mechanism evidence.

Note on ε. Both CRM always-on and CRM τ-gated in Table 1 use ε=0.2, matching the τ-sweep configuration. The K-matched design ensures the only variable is conformal gating (τ→0 vs. calibrated τ).

C Uncertainty Proxy Comparison

Table 6: Spearman ρ between candidate uncertainty proxies and K=10 action variance (ground-truth uncertainty). Evaluated on LIBERO-Object (10 tasks × 10 rollouts, 27,626 steps). Only pairwise L_2 distance achieves meaningful correlation.

Proxy	Spearman ρ	Extra VLA calls
K=2 pairwise L_2 (ours)	**0.9979**	1
K=1 CRM correction magnitude	0.152	0
K=2 CRM correction magnitude	0.134	1
Flow velocity dispersion (FMVD)	0.083	0

Table 6 compares four candidate uncertainty proxies against K=10 sample variance as ground truth. CRM correction magnitude (the L_2 norm of the CRM's predicted residual) is attractive because it requires no extra VLA forward passes, but correlates weakly with true uncertainty ($\rho < 0.16$): the CRM predicts similar-magnitude corrections regardless of whether the VLA is confident or

uncertain. FMVD, which measures dispersion in the flow-matching velocity field, performs worse still (ρ=0.083). The K=2 pairwise L_2 distance achieves near-perfect correlation (ρ=0.9979) at the cost of one additional VLA forward pass, which is absorbed by batched inference (Section 4.4).

D Reproducibility Details

Hardware. All experiments run on NVIDIA RTX 4090 GPUs (24 GB VRAM). SmolVLA inference uses a single GPU; CRM training and evaluation each use a single GPU.

CRM training. The CRM is trained on three suites (Spatial, Object, Goal; 30 tasks, ∼88K samples) for 50 epochs with Adam (lr=10^{-3}, batch size 32) in ∼6 GPU-hours on 1× RTX 4090. Long is excluded from CRM training; Long-suite evaluation tests cross-suite generalization. Precomputing the K=10 forward-pass cache (required once per suite) takes ∼2 GPU-hours per suite.

Evaluation. Each suite evaluation (20 rollouts × 10 tasks × 300 max steps) takes ∼2.5 GPU-hours. A full 5-method × 4-suite × 3-seed experiment requires ∼150 GPU-hours total.

Software. PyTorch 2.4, Transformers 4.45, LIBERO (MuJoCo-based). Code and CRM checkpoints will be released upon acceptance.

E Extended 8-Seed Analysis on Spatial

Table 7: Per-seed success rates (%) on LIBERO-Spatial for VLA-only vs. CRM τ-gated (K=2, τ=0.25). All 8 seeds use identical K=2 configuration (code-verified).

Seed	VLA-only	CRM τ-gated	Δ
42	80.0	82.0	+2.0
43	79.0	80.0	+1.0
44	78.5	80.5	+2.0
45	80.0	80.0	+0.0
46	80.5	80.0	−0.5
47	80.0	81.5	+1.5
48	79.5	81.0	+1.5
49	78.5	82.0	+3.5
Mean±std	79.5 ± 0.8	80.9 ± 0.9	+1.4

Table 7 reports per-seed results for the extended 8-seed paired analysis on LIBERO-Spatial. A paired t-test yields Δ=+1.4 pp, p=0.017 (n=8), reaching statistical significance at α=0.05. Six of eight seeds show positive Δ; the two non-positive seeds (Δ=0.0 and Δ=−0.5) are consistent with measurement noise rather than systematic degradation. This analysis provides the strongest statistical evidence that conformal τ-gating produces a genuine improvement on the suite where the uncertainty gate fires most frequently (trigger rate ∼20%).

F Additional Ablations

Table 8 reports two ablations that isolate multi-sample selection without learned correction. K=10 argmin invests 10× the VLA forward passes but hurts on Goal (−2.9 pp) and Long (−1.5 pp): selection alone cannot fix systematic errors. K=2 mean offers no learned correction mechanism (−0.7 pp Goal, −1.0 pp Long; Long is single-seed preliminary), confirming that multi-sampling alone is insufficient.

Table 8: Supplementary ablations on SmolVLA. K=10 argmin [8]: argmin-to-centroid selection over 10 samples, no learned correction. K=2 mean: centroid of two samples, no CRM. Neither consistently improves over VLA-only.

Method	K	Spatial (%)	Object (%)	Goal (%)	Long (%)
VLA-only	1	79.3 ± 0.8	77.7 ± 1.3	83.7 ± 1.3	58.0 ± 1.5
K=10 argmin	10	80.3 ± 1.0	77.3 ± 1.5	80.8 ± 0.3	$56.5^{\dagger}$
K=2 mean	2	—	77.5 ± 2.3	83.0 ± 2.2	$57.0^{\dagger}$

n=3 seeds unless noted; $^{\dagger}n$=1; — = omitted (computational budget).

Table 9: Per-seed success rates (%) for CRM always-on (K=2, ε=0.2, $\tau \to 0$). SmolVLA uses seeds 42–46 (n=5); Table 1 reports n=3 means from seeds 42–44. $\pi_{0.5}$ results are preliminary (limited seeds).

Model	Suite	s42	s43	s44	s45	s46	n	Mean±Std
SmolVLA	Spatial	79.0	76.0	79.0	77.0	78.5	5	77.9 ± 1.3
SmolVLA	Object	71.0	73.5	74.5	74.5	67.5	5	72.2 ± 3.0
SmolVLA	Goal	80.0	79.5	80.5	82.0	80.0	5	80.4 ± 1.0
SmolVLA	Long	41.0	45.5	43.0	—	—	3	43.2 ± 2.3
$\pi_{0.5}$	Spatial	96.0	—	—	—	—	1	96.0
$\pi_{0.5}$	Object	90.5	92.0	—	—	—	2	91.2 ± 1.1
$\pi_{0.5}$	Goal	81.5	—	—	—	—	1	81.5
$\pi_{0.5}$	Long	77.0	—	—	—	—	1	77.0

—: not evaluated. SmolVLA Long s45 incomplete (excluded).

G CRM Always-On Per-Seed Analysis

Table 9 reports per-seed CRM always-on results. SmolVLA means over seeds 42–44 match Table 1; the extended 5-seed means show consistent degradation patterns across all suites. Object exhibits the highest per-seed variance (std=3.0), driven by seed 46 (67.5%), suggesting interaction between CRM noise injection and task-specific stochasticity.

H $\pi_{0.5}$ Extended Multi-Seed Results

Table 10: $\pi_{0.5}$ (3.6B) per-seed success rates (%) across three methods and four LIBERO suites from dedicated multi-seed evaluation runs. Table 3 reports seed-42 matched pairs from separate runs; per-seed differences ($\leq$4 pp) reflect cross-server variance and evaluation stochasticity.

Suite	Method	s42	s43	s44	n	Mean±Std
Spatial	VLA-only	98.0	97.5	98.5	3	98.0 ± 0.5
Spatial	CRM always-on	96.5	97.0	97.0	3	96.8 ± 0.3
Spatial	CRM τ-gated	96.5	98.0	97.0	3	97.2 ± 0.8
Object	VLA-only	100.0	98.5	99.5	3	99.3 ± 0.8
Object	CRM always-on	89.0	87.0	88.0	3	88.0 ± 1.0
Object	CRM τ-gated	85.5	85.0	88.0	3	86.2 ± 1.6
Goal	VLA-only	98.0	97.0	97.5	3	97.5 ± 0.5
Goal	CRM always-on	85.5	83.0	85.0	3	84.5 ± 1.3
Goal	CRM τ-gated	89.5	89.5	—	2	89.5 ± 0.0
Long	VLA-only	94.0	95.5	95.5	3	95.0 ± 0.9
Long	CRM always-on	—	—	76.0	1	76.0
Long	CRM τ-gated	80.5	80.5	83.0	3	81.3 ± 1.4

—: not evaluated. VLA-only and CRM evaluated on different servers (seed-matched).

Table 10 extends the preliminary single-seed results in Table 3 to multi-seed evaluation on $\pi_{0.5}$. Unconditional CRM correction degrades all four suites relative to VLA-only, confirming the single-

seed pattern. On Object, CRM τ-gated (86.2%) performs below CRM always-on (88.0%), indicating that gating effectiveness depends on the quality of the underlying CRM corrections, which diminishes when the base policy is near-optimal (VLA-only 99.3%). On Long, τ-gating recovers 5.3 pp relative to CRM always-on (81.3 vs. 76.0), consistent with OOD suppression. Goal shows partial recovery: τ-gating (89.5) narrows the gap to VLA-only (97.5) relative to CRM always-on (84.5).

I K-Sweep Analysis

Table 11: Effect of sample count K on CRM always-on performance (SmolVLA, LIBERO-Object, ε=0.2). Increasing K beyond 2 provides marginal improvement; K=10 yields the highest SR but at 5× the VLA cost.

K	VLA calls	Seeds	SR (Mean±Std)	Δ vs. K=2
2	2	5	72.2 ± 3.0	—
5	5	2	73.2 ± 0.4	+1.0
10	10	1	78.0	+5.8
20	20	2	74.5 ± 1.4	+2.3

K=2: from Table 9 (n=5). K=10 uses ε=0.05 (single seed).

Table 11 varies the sample count K for CRM always-on on LIBERO-Object. Performance is relatively stable across K=2–20 (72.2–78.0%), with K=10 showing the highest point estimate but at 5× the VLA forward-pass cost and with only a single seed. The marginal gain from larger K does not justify the computational cost, supporting the paper's use of K=2 with conformal gating as the preferred operating point.